Code No: **20CS5T02/20IT5T02**

R20

III B. TECH I SEMESTER REGULAR EXAMINATIONS, NOVEMBER - 2022
ARTIFICIAL INTELLIGENCE
(Common to CSE and INF)

Time: 3 Hours **Max. Marks: 70**

Note : Answer **ONE** question from each unit **(5 × 14 = 70 Marks)**

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UNIT-I

1. a) Define Artificial Intelligence. Explain the techniques of A.I. Also describe the characteristics of Artificial Intelligence. [7M]

   b) List and explain the applications of Artificial Intelligence. [7M]

(OR)

2. a) Explain about problem reduction with an example. [4M]

   b) Discuss Constrain Satisfaction Problem with the help of [10M]

   CROSS

   + ROADS

   DANGER

UNIT-II

3. a) Describe the alpha-beta Pruning algorithm with an example. [7M]

   b) What is predicate logic? Explain the predicate logic representation with suitable example. [7M]

(OR)

4. a) Solve the water-jug problem by writing the production rules. [7M]

   b) Give a brief note on Axiomatic System. [7M]

UNIT-III

5. a) Give semantic nets to describe the following: [7M]
   Narayan is a writer
   Narayan lives in Bombay
   Ishwar is a teacher
   Ishwar lives in Bangalore.
   Narayan sent a copy of his book to Ishwar
   Ishwar sent his thanks to Narayan.

   b) What is the significance of knowledge representation? Give differences between database and knowledge base. [7M]
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(OR)

6.	a)	Provide relational structures for slot and filler structures. Compare their merits and demerits.	[7M]
	b)	Suggest a semantic network to describe the furniture in a house.	[7M]

UNIT-IV

7.	a)	List out and explain the characteristics features of expert system.	[7M]
	b)	Define certainty factor. What are the components of certainty factor?	[7M]

(OR)

8.	a)	Discuss the Bayesian Belief networks with an example.	[10M]
	b)	Discuss in brief about Dempster-Shafer theory.	[4M]

UNIT-V

9.	a)	Discuss Membership functions in detail.	[7M]
	b)	Explain inference rules in fuzzy logic.	[7M]

(OR)

10.	a)	Differentiate variables and hedges.	[7M]
	b)	Discuss the architecture of fuzzy systems.	[7M]

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